WORKSHEET 5.9

Moving Between the Graphs of f, f' and f''

Unit 5: Analysing Functions

Key ideas

- Where the graph of f' is **above** the axis, f is increasing; below, decreasing.
- f has a relative maximum where f' crosses the axis from above to below, and a minimum where it crosses from below to above.
- Where the graph of f' is **rising**, f is concave up; falling, concave down.
- f has a point of inflection where f' has a relative maximum or minimum. Always say which graph you are describing.

Worked examples

Example 1 · Reading the derivative

The graph of f' consists of the line segments joining $(0, -2)$, $(2, 2)$ and $(5, -1)$. Find the relative extrema of f.

Solution. f' crosses the axis at $x = 1$ (from negative to positive) and at $x = 4$ (from positive to negative).

Relative minimum at $x = 1$; relative maximum at $x = 4$.

Example 2 · Concavity from the derivative

For the same f', where is the graph of f concave up, and where is its point of inflection?

Solution. f' is increasing on $(0, 2)$ and decreasing on $(2, 5)$.

Concave up on $(0, 2)$; point of inflection at $x = 2$.

Example 3 · From f to f prime

On an interval, f is increasing and concave down. What is true of f' there?

Solution. Increasing: $f' > 0$. Concave down: f' is decreasing.

f' is positive and decreasing.

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Practice

Questions 5 to 8 refer to a function f that is continuous on $[-2, 6]$. The graph of its derivative f' consists of three line segments joining the points $(-2, 4)$, $(1, -2)$, $(3, 2)$ and $(6, -1)$, in that order.

CORE

1. If $f'(x) > 0$ on $(1, 5)$, what is true of f on that interval?

2. If f' has a relative maximum at $x = 3$, what does the graph of f have at $x = 3$?

3. If $f''(x) < 0$ on $(0, 2)$, what is true of f' and of the graph of f there?

4. If $f'(2) = 0$ and f' changes from positive to negative at $x = 2$, what does f have there?

EXAM LEVEL

5. Find the x-coordinates of the relative maxima of f. Justify your answer.

6. Find the x-coordinate of the relative minimum of f. Justify your answer.

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7. On what interval is the graph of f concave up? Where are its points of inflection?

8. Find $f''(2)$ and $f''(4)$.

Question 9 · Free-response practice

The derivative of a function f is $f'(x) = x^2 - 4x + 3$.

(a) On what intervals is f increasing? Justify your answer.

(b) Find the x-coordinates of the relative maximum and the relative minimum of f.

(c) Find the x-coordinate of the point of inflection of the graph of f, and explain how it can be seen on the graph of f'.

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Worked solutions

1. f is increasing on $(1, 5)$.
2. A point of inflection.
3. f' is decreasing and the graph of f is concave down.
4. A relative maximum.
5. $x = 0$ and $x = 5$: f' changes from positive to negative at each.
6. $x = 2$: f' changes from negative to positive.
7. Concave up on $(1, 3)$, where f' is increasing. Points of inflection at $x = 1$ and $x = 3$, where f' changes direction.
8. The slopes of the graph of f': $f''(2) = \dfrac{2-(-2)}{3-1} = 2$ and $f''(4) = \dfrac{-1-2}{6-3} = -1$.
9. (a) $f'(x) = (x-1)(x-3) > 0$ for $x < 1$ and for $x > 3$.
   (b) Maximum at $x = 1$ (f' changes from positive to negative); minimum at $x = 3$.
   (c) $f''(x) = 2x - 4$ changes sign at $x = 2$. On the graph of f' this is the vertex of the parabola, where f' changes from decreasing to increasing.

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